Parametric Equations & Modeling

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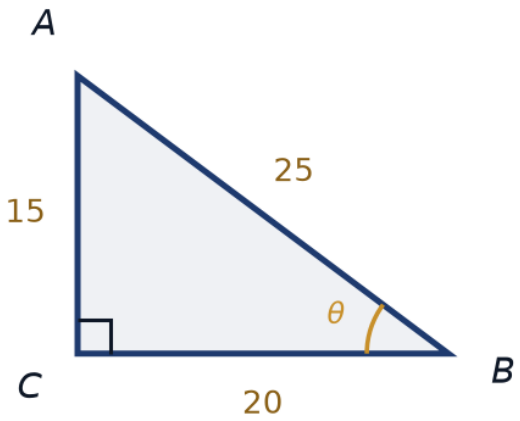
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Evaluating parametric equations

Given $x(t) = t^2 - 1$ and $y(t) = 2t + 3$, find the point on the curve at $t = 2$.

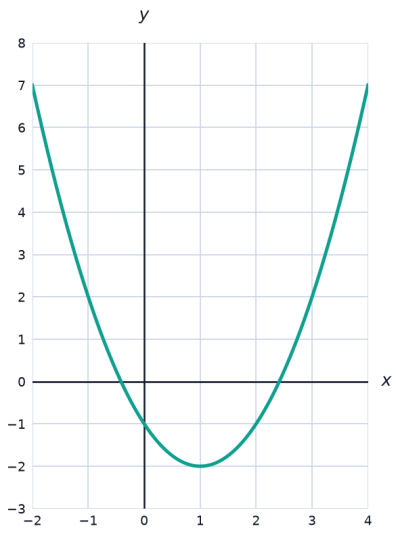
2 [Foundational] • Target: Evaluating parametric equations

Given $x(t) = 3\cos t$ and $y(t) = 3\sin t$, find the point at $t = \frac{\pi}{3}$, and identify the shape traced by the curve.



3 [Foundational] • Target: Eliminating the parameter

Eliminate the parameter from $x(t) = t + 1$, $y(t) = t^2 - 2$ to find the rectangular equation, and identify the curve.



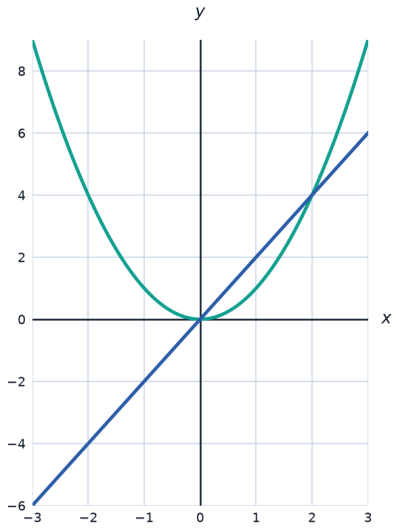
4 [Foundational] • Target: Eliminating the parameter

Eliminate the parameter from $x(t) = 2t - 3$, $y(t) = 4t + 1$ to find the rectangular equation.

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5 [Foundational] • Target: Visualizing a parametric curve's components

The graph shows the two component functions $x(t) = t^2$ (upper curve) and $y(t) = 2t$ (lower, straight line) for the parametric curve $x(t) = t^2$, $y(t) = 2t$, plotted separately against t on the horizontal axis.

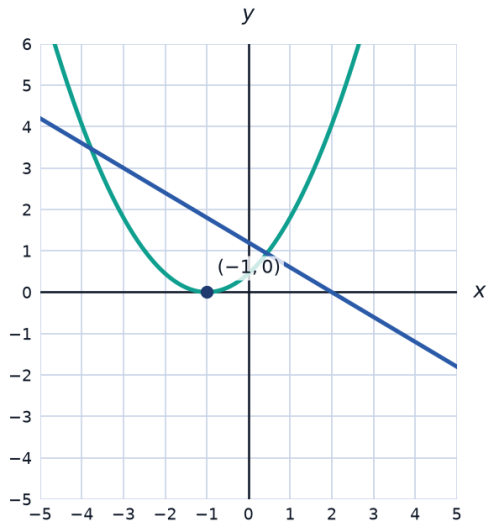


a Eliminate the parameter t to find the rectangular equation relating x and y for this curve.

b Use your rectangular equation to describe the shape of the actual parametric curve (not the component graphs shown).

6 [Foundational] • Target: Parametric equations for lines and conics

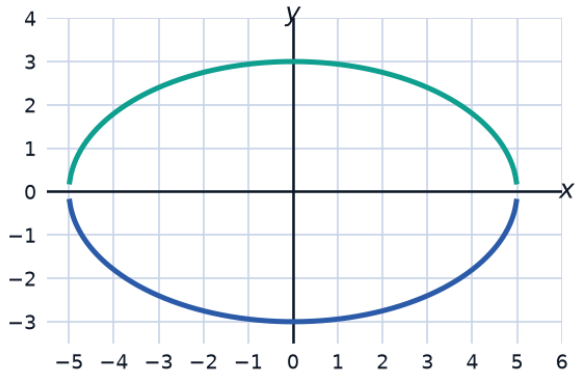
Write parametric equations for the line segment from $(1, 2)$ to $(5, 10)$, using $t \in [0, 1]$.



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7 [Foundational] • Target: Parametric equations for lines and conics

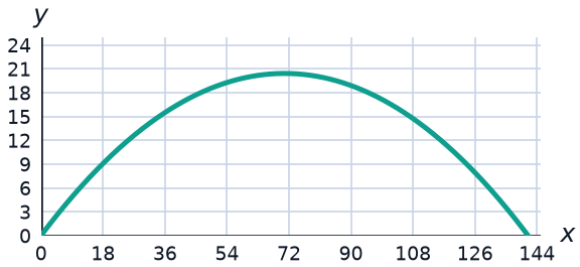
Write parametric equations for an ellipse with semi-major axis 5 (horizontal) and semi-minor axis 3 (vertical), centered at the origin.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Modeling projectile motion parametrically

A ball is launched with initial speed 40 m/s at an angle of 30° above horizontal. Its position is modeled by $x(t) = 40\cos(30^\circ)\,t$ and $y(t) = 40\sin(30^\circ)\,t - 4.9t^2$. Find the ball's position at $t = 2$ seconds, to one decimal place.



9 [Intermediate] • Target: Modeling projectile motion parametrically

For the same projectile, find the time at which the ball lands (returns to $y = 0$).

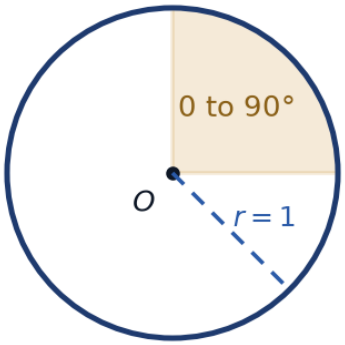
10 [Intermediate] • Target: Comparing parametric speed and direction

Two objects follow the same path $y = x^2$, but object A uses $x(t) = t$, $y(t) = t^2$ while object B uses $x(t) = 2t$, $y(t) = 4t^2$. Explain why they trace the same curve but do not move through it at the same rate.

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11 [Intermediate] • Target: Restricting the parameter's domain

The parametric equations $x(t) = \cos t$, $y(t) = \sin t$ trace the full unit circle for $t \in [0, 2\pi)$. Describe the portion of the circle traced when the domain is restricted to $t \in [0, \frac{\pi}{2}]$.

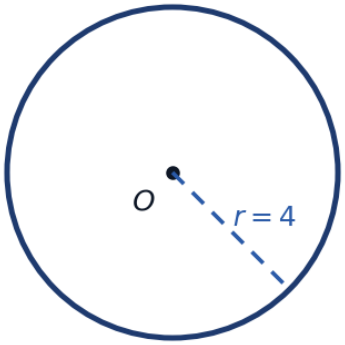


12 [Intermediate] • Target: Restricting the parameter's domain

Explain why restricting t in $x(t) = t^2$, $y(t) = t$ to $t \geq 0$ changes the rectangular relation from $x = y^2$ (a full sideways parabola) to only its upper half.

13 [Intermediate] • Target: Parametric equations for circular motion with a phase shift

A point moves along a circle of radius 4 according to $x(t) = 4\cos\left(t + \frac{\pi}{4}\right)$, $y(t) = 4\sin\left(t + \frac{\pi}{4}\right)$. Find its position at $t = 0$, and explain the effect of the $\frac{\pi}{4}$ phase shift compared to $x(t) = 4\cos t$, $y(t) = 4\sin t$.



14 [Intermediate] • Target: Comparing parametric speed and direction

Eliminate the parameter to determine the Cartesian equation of the curve $x = 1 + 2\sec t$, $y = -2 + 3\tan t$. State the shape, center, and vertices.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Restricting the parameter's domain • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Given vectors $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = -3\mathbf{i} + 4\mathbf{j}$, find the scalar projection of $\mathbf{a}$ onto $\mathbf{b}$ and the angle θ between them to the nearest tenth of a degree.

16 [Challenging] • Target: Parametric equations for circular motion with a phase shift • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

[Vectors, Lines & Parametric Kinematics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

17 [Challenging] • Target: Evaluating parametric equations • Cross-Lesson: 4.01 Inverse & Composite Functions

Decompose the vector $\mathbf{u} = \langle 6, 2 \rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4 \rangle$.

18 [Challenging] • Target: Eliminating the parameter • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

A right circular cone has a base radius of 6 cm and a slant height of 10 cm. Find (a) its perpendicular vertical height h, (b) its volume V in terms of π, and (c) its total surface area in terms of π.

19 [Challenging] • Target: Visualizing a parametric curve's components • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Find all solutions to the trigonometric equation $2\sin^2 x - 3\sin x + 1 = 0$ on the domain $x \in [0, 2\pi)$.

20 [Challenging] • Target: Parametric equations for lines and conics • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Find the volume of a right circular cone with base radius 6 and height 8.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Evaluating parametric equations

Eliminate the parameter from $x(t) = t - 2$ and $y(t) = -3t + 4$ to find the rectangular equation for y in terms of x.

a. $y = -3x - 1$

b. $y = -3x + 4$

c. $y = 3x - 2$

d. $y = -3x - 2$

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22 [Foundational] • Target: Eliminating the parameter

A curve is given by $x(t) = 4\cos t$, $y(t) = 4\sin t$. Find the exact point on the curve at $t = \frac{11\pi}{6}$ (330°).

a. $(2\sqrt{3}, 2)$

b. $(2\sqrt{3}, -2)$

c. $(-2, 2\sqrt{3})$

d. $(-2\sqrt{3}, -2)$

23 [Foundational] • Target: Visualizing a parametric curve's components

A projectile is launched with initial speed 34 m/s at 30° above horizontal, so $y(t) = 34\sin(30^\circ)\,t - 4.9t^2$. Find $y(2)$, to one decimal place.

a. 12.4

b. 19.4

c. 16.4

d. 14.4

24 [Foundational] • Target: Parametric equations for lines and conics

Two particles, A and B, move in the xy-plane. Their position vectors at time t seconds, for $t \geq 0$, are given by $\mathbf{r}_A(t) = (2t;\, 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t;\, 1 - 2t)$ respectively. The particles start their motion at $t = 0$. Find the minimum distance between particle A and particle B.

a. $\sqrt{26}$

b. $2\sqrt{13}$

c. $3\sqrt{10}$

d. $4\sqrt{5}$

25 [Foundational] • Target: Modeling projectile motion parametrically

Consider two vectors, $\mathbf{u} = (3;\, k)$ and $\mathbf{v} = (-2;\, 4)$. If $\mathbf{u}$ is parallel to $\mathbf{v}$, find the value of the scalar parameter k.

a. -6

b. 6

c. 3

d. -3

26 [Foundational] • Target: Comparing parametric speed and direction

Two vectors are given by $\vec{u} = 3\hat{i} + k\hat{j}$ and $\vec{v} = -\hat{i} + 4\hat{j}$. If $\vec{u}$ and $\vec{v}$ are orthogonal, find the value of the scalar parameter k.

a. 3/4

b. 4/3

c. -3/4

d. -4/3

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27 [Foundational] • Target: Restricting the parameter's domain

Consider the vector $\mathbf{v}$ with initial point $P = (2, 3)$ and terminal point $Q = (5, 7)$. What are the components of the vector $\mathbf{v}$?

a. ■3, 4■

b. ■7, 10■

c. ■-3, -4■

d. ■4, 3■

28 [Foundational] • Target: Parametric equations for circular motion with a phase shift

Find the unit vector in the direction of $\mathbf{v} = \langle -8, 15 \rangle$.

a. $\left\langle \frac{9}{17}, \frac{15}{17} \right\rangle$

b. $\left\langle \frac{15}{17}, -\frac{8}{17} \right\rangle$

c. $\left\langle -\frac{8}{17}, \frac{15}{17} \right\rangle$

d. $\langle -8, 15 \rangle$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Evaluating parametric equations

In the standard (x, y) coordinate plane, the lines with equations $2x - y = 7$ and $x + y = 5$ intersect at the point (a, b). What is the value of a?

a. 1

b. 3

c. 4

d. 5

30 [Intermediate] • Target: Eliminating the parameter

Consider two points in the Cartesian plane, $P = (2, 3)$ and $Q = (5, 7)$. What is the component form of the vector $\vec{PQ}$?

a. $\langle 3, 4 \rangle$

b. $\langle -3, -4 \rangle$

c. $\langle 7, 10 \rangle$

d. $\langle 1, 4 \rangle$

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31 [Intermediate] • Target: Visualizing a parametric curve's components

Consider the vector $\mathbf{v} = \langle 3, 4 \rangle$. What is the magnitude of this vector?

a. 5

b. 7

c. 12

d. 25

32 [Intermediate] • Target: Parametric equations for lines and conics

Two lines have slopes $m_1 = -1$, $m_2 = -1$ and y-intercepts $b_1 = -2$, $b_2 = -2$. Classify the system.

a. Consistent and independent (one solution)

b. Consistent and dependent (infinitely many solutions)

c. Inconsistent (no solution)

d. Cannot be determined

33 [Intermediate] • Target: Modeling projectile motion parametrically

Find the projection of $\mathbf{u} = \langle 3, 2 \rangle$ onto $\mathbf{v} = \langle 2, 2 \rangle$.

a. $\left\langle \frac{3}{2}, \frac{5}{2} \right\rangle$

b. $\left\langle \frac{5}{2}, \frac{5}{2} \right\rangle$

c. $\left\langle \frac{7}{2}, \frac{5}{2} \right\rangle$

d. $\frac{5}{4}$

34 [Intermediate] • Target: Comparing parametric speed and direction

Find the angle between $\mathbf{u} = \langle 2, 1 \rangle$ and $\mathbf{v} = \langle 4, 2 \rangle$ (to one decimal place).

a. 0.0°

b. 20.0°

c. 180.0°

d. 10.0°

35 [Intermediate] • Target: Restricting the parameter's domain

Find the parametric equations for the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

a. $x = 4\sin t, y = 3\cos t$

b. $x = 4t, y = 3t$

c. $x = 4\cos t, y = 3\sin t$

d. $x = \cos t, y = \sin t$

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36 [Intermediate] • Target: Parametric equations for circular motion with a phase shift

For the hyperbola $\frac{x^2}{4} - \frac{y^2}{9} = 1$, what are the equations of the asymptotes?

a. $y = \pm\frac{3}{2}x$

b. $y = \pm\frac{2}{3}x$

c. $y = \pm 6x$

d. $y = \pm\frac{5}{2}x$

37 [Intermediate] • Target: Evaluating parametric equations

Find the period and phase shift of $y = \tan(1/2x - 1)$.

a. Period 2π, phase shift 0

b. Period 2π, phase shift 2 units left

c. Period π, phase shift 2 units right

d. Period 2π, phase shift 2 units right

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Eliminating the parameter • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Graph $y = 2\sin(3x - \pi)$: state amplitude, period, phase shift.

a. Amp 2, period 2π, phase π

b. Amp 2, period π

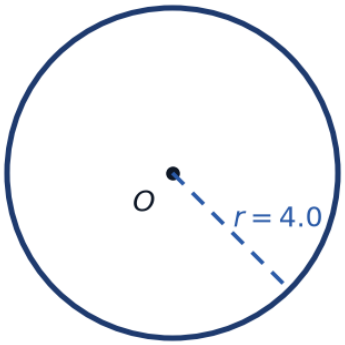
c. Amp 3, period $2\pi/3$

d. Amplitude 2, period $2\pi/3$, phase shift $\pi/3$ right

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39 [Challenging] • Target: Visualizing a parametric curve's components • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

What is the center of the circle $(x-3)^2+(y+2)^2=16$?



a. (-3, 2)

b. (3, -2)

c. (3, 2)

d. (-2, 3)

40 [Challenging] • Target: Parametric equations for lines and conics • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Let m be a real parameter. Consider the quadratic equation $z^2-2mz+(m^2+3m-6)=0$ with complex roots z_1 and z_2. In the complex plane, the origin $O(0)$ and the points representing z_1 and z_2 form the vertices of a non-degenerate equilateral triangle. What is the sum of the areas of all possible equilateral triangles that can be formed in this way?

a. $15\sqrt{3}$

b. $12\sqrt{3}$

c. $\frac{15\sqrt{3}}{2}$

d. $9\sqrt{3}$

41 [Challenging] • Target: Modeling projectile motion parametrically • Cross-Lesson: 1.03 Rates of Change & Concavity

Consider the system of linear equations represented by the matrix equation $(1,1,1;1,2,3;3,5,k)(x;y;z)=(6;14;m)$ where k and m are real constants. Suppose the system is consistent and has infinitely many solutions. Let $P(x,y,z)$ be the unique solution that minimizes the squared Euclidean distance from the origin, $d^2=x^2+y^2+z^2$. What is the value of $k+m+d^2$?

a. 55

b. 47

c. 38

d. 109

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42 [Challenging] • Target: Comparing parametric speed and direction • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Consider the system of linear equations below: $\{3x - 2y = 8; kx + 6y = 5$ For which of the following values of the constant k does the system have no solution (x, y)?

a. -9

b. -4

c. 4

d. 9

43 [Challenging] • Target: Restricting the parameter's domain • Cross-Lesson: 2.02 Logarithmic Functions & Properties

If the real numbers x and y satisfy the system of equations $x^3 - 3xy^2, = 10; 3x^2y - y^3, = 5$ what is the value of $x^2 + y^2$?

a. $\sqrt{5}$

b. 5

c. $5\sqrt{5}$

d. 25

44 [Challenging] • Target: Parametric equations for circular motion with a phase shift • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

In the system of linear equations below, k is a real constant. $(k - 1)x + 6y, = 18; 2x + ky, = 12$ For which value of k does the system have infinitely many solutions (x, y)?

a. -4

b. -3

c. 3

d. 4

45 [Challenging] • Target: Evaluating parametric equations • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

In the standard (x, y) coordinate plane, the system of equations $\{y = |x^2 - 4|; y = 2x + k$ has exactly 3 distinct real solutions (x, y). What is the sum of all possible values of the real constant k?

a. 4

b. 5

c. 9

d. 11

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